Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/10/2011
PI Name:	Redpath, John Leslie Ph.D.		
Project Title:	High energy proton dose-rate and mixed field effects on neoplastic transformation in vitro		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jlredpat@uci.edu	**Fax:**	FY (949) 824-3566
PI Organization Type:	UNIVERSITY	**Phone:**	(949) 824-7395
Organization Name:	University of California Irvine		
PI Address 1:	Radiation Oncology		
PI Address 2:	School of Medicine, Med Sci I, B148		
PI Web Page:			
City:	Irvine	**State:**	CA
Zip Code:	92697	**Congressional District:**	48
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	01/01/2007	**End Date:**	12/31/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received extension to 12/31/2011 (from 12/31/2010) per C. Guidry/JSC (Ed., 2/3/11)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Elmore, Eugene (University of California Irvine)		
Grant/Contract No.:	NNJ06HD90G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The central objectives are 1) to determine the the factor by which the effectiveness of high energy protons is reduced when they are delivered at dose-rates that correspond to solar particle events, and 2) to determine the effectiveness of mixed fields of protons and heavy ions compared to either alone. The cancer-relevant endpoint of neoplastic transformation in vitro will be used. The work is of relevance to the improvement of estimation of cancer risks to astronauts involved in space exploration and address certain emphases in the Bioastronautics Roadmap.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Provides greater insight into cancer risk from exposure to low doses of high energy proton and iron ion irradiation.
Task Progress:	During the reporting period we completed the proposed studies on mixed beam effects and proton dose-rate effects using the endpoint of neoplastic transformation in vitro. These studies were then written up and submitted for publication in early 2011. These manuscripts have subsequently been accepted for publication. The salient findings during this period are: 1. Confirmation of our earlier reported threshold-type dose-response curve for 1 GeV/n iron ions. This has been attributed to an adaptive response induced by the associated low LET delta rays. 2. In mixed beam studies of iron ions and protons with either a 15 min or 24 h interval between beams there was no evidence of any adaptation by the initial beam agianst the effects of the second beam. The data indicated an essentially additive interaction. Subsequent experiments indicated that the failure to see any evidence of an adaptive response may well have been due to the time intervals chosen and that an 8 h interval may show adaptation. This would have to be verified by further experimentation. 3. The proton dose-rate studies comparing 20 cGy/h to 50 cGy/min over the range 5-100 cGy showed no dose-rate effect. 4. Comparison of dose -response curves for 232 MeV/n protons (Loma Linda) with 1 GeV/n protons (NSRL) showed similar effects up to about 50 cGy. The data indicated that at 100 cGy the 1 GeV/n protons may be more biologically effective but repeat experiments would ahve to be done to confirm this observation.
Bibliography Type:	Description: (Last Updated: 01/13/2012)
Articles in Peer-reviewed Journals	Elmore E, Lao X-Y, Kapadia R, Redpath JL. "Effect of dose-rate on induction of neoplastic transformation in vitro by low doses of 232 MeV protons." Int J Radiat Biol. 2011 Jun;87(6):622-7. PMID: 21627566 , Jun-2011
Articles in Peer-reviewed Journals	Elmore E, Lao X-Y, Kapadia R, Swete M, Redpath JL. "Neoplastic transformation by mixed beams of high energy iron ions and protons." Radiation Research, accepted for publication, June 2011. , Jun-2011